



Utilization of Banked Human Adipose Allograft for Host Mediated Replacement of Dermal White Adipose Tissue and Volumization in Plastic and Reconstructive Surgery: A Systematic Review and Meta-Analysis

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Abstract

Background: Autologous Fat Grafting (AFG) remains the reference standard for soft-tissue and Dermal White Adipose Tissue (dWAT) volumization, but donor-site morbidity and unpredictable graft retention (26-83%) limit its reliability. 1,2 Banked human adipose tissue allografts, processed into an acellular or minimally manipulated Allograft Adipose Matrix (AAM), have been proposed as an off-the-shelf, host-remodeled alternative that supports neo adipogenesis without a donor-site procedure [1-4].

Objective: To examine Quantificare-validated results on 35 consecutive patients injected with a human adipose tissue allograft, (Britecyte, Frederick, MD), and to systematically review and, meta-analyze human clinical outcomes of banked adipose tissue allograft used for dermal/subcutaneous adipose replacement and soft-tissue volumization in plastic and reconstructive surgery.

Methods: Under the International Cell Surgical Society (IRB, ICSS -2021-011), from 09/2025-08/2026, 35 patients were treated with a Human Adipose Tissue Allograft (hATA), (Lipoderma, Britecyte, Inc, Frederick MD. Quantificare (Paris, France) 3-D and Volumetric analysis was performed monthly, on all patients. A systematic review was conducted following PRISMA 2020 guidelines [5]. PubMed/MEDLINE, Embase, Cochrane CENTRAL, and Scopus were searched through June 2026, supplemented by ClinicalTrials.gov and citation tracking. Human studies reporting clinical, histologic, or patient-reported outcomes after banked adipose allograft injection or implantation were included; animal-only and in-vitro-only studies were excluded from quantitative synthesis. Data on volume retention, responder/satisfaction rate, and adverse events were extracted and pooled using random-effects (DerSimonian-Laird) proportion and mean-difference models where at least two comparable datasets existed [6].

Results: Fourteen human studies (approximately 96 patients) met eligibility criteria for qualitative synthesis; four comparable datasets from three prospective studies were suitable for quantitative pooling. The pooled clinical responder/satisfaction rate was 77.2% (95% CI, 59.4-95.0%; $I^2 = 0\%$; 2 studies, 21 patients). Pooled midface volumetric retention (malar and prejowl subsites combined) at 24 weeks was 23.9% (95% CI, 15.8-31.9%; $I^2 = 0\%$), while temple injection was associated with a separately reported, non-pooled retention of approximately 75% at 6 months. Across all included studies, adverse events were limited to self-resolving injection-site erythema, swelling, and discomfort; no graft loss, infection, nodularity, or immune rejection was reported. Histologic studies demonstrated progressive host-cell infiltration, neovascularization, and perilipin-positive adipocyte formation within the

graft scaffold by 3-6 months [3,4]. This data was supported by our Clinical Study, which showed skin quality improvement, and volumization, mirroring AFG results with repeated sessions, two, spaced every 3 months with maximal patient satisfaction seen at one year with host-mediated tissue replacement.

Conclusions: Banked human adipose tissue allograft is a biologically plausible and clinically well-tolerated option for host-mediated replacement of dWAT and soft-tissue volumization, with a favorable safety profile. Modest and anatomically variable volumetric retention is seen with the initial injection, with increases occurring with recipient adipose integration and multiple sessions. The evidence base remains limited to small, non-randomized studies; larger comparative and randomized trials are needed before this technology can be considered an established substitute for autologous fat grafting.

Keywords: Adipose Allograft; Dermal White Adipose Tissue; Fat Grafting; Regenerative Medicine; Tissue Engineering; Volumization

Introduction

Dermal White Adipose Tissue (dWAT) is now recognized as a histologically and functionally distinct adipocyte compartment within the reticular dermis, separate from Subcutaneous White Adipose Tissue (sWAT), that contributes to thermoregulation, hair-follicle cycling, innate immune defense, wound healing, and mechanical cushioning of the skin [7-12]. Loss of dermal and subcutaneous adipose volume—whether from intrinsic aging, photodamage, GLP-1 associated weight loss, surgical excision, trauma, lipodystrophy, or oncologic resection—produces visible contour deformity, skin laxity, and, in load-bearing regions such as the plantar foot, loss of protective cushioning that predisposes to ulceration [7,9,11,12]. Because adipocytes reach replicative and hypertrophic stasis after puberty and do not meaningfully self-renew once destroyed, damaged adipose tissue represents a largely irreversible deficit without intervention [3]. Autologous Fat Grafting (AFG), first described over a century ago and refined by Coleman's structural fat grafting technique, remains the most widely used method for restoring soft-tissue and dermal volume in aesthetic and reconstructive surgery [13]. AFG is attractive because it transfers living, vascularizable autologous tissue, but its clinical utility is constrained by several well-documented limitations: donor-site morbidity, operative time required for harvest and processing, and highly variable graft survival, with reported long-term volumetric retention ranging from approximately 26% to 83% depending on anatomic site, technique, and patient factors [14-16].

A 2018 systematic review and meta-analysis of AFG in facial reconstruction similarly reported wide heterogeneity in retention and complication rates across the published literature, underscoring the need for a more standardized, reproducible alternative [16]. Banked (allogeneic) human adipose tissue offers a potential solution to these limitations. Adipose tissue obtained from screened, deceased tissue donors through tissue banks

accredited by the American Association Of Tissue Banks (AATB) and regulated under the U.S. Food and Drug Administration's Good Tissue Practice framework (21 CFR Part 1271) can be processed—typically through detergent-based delipidization, decellularization, or minimal-manipulation protocols—into an off-the-shelf Allograft Adipose Matrix (AAM) that preserves the native adipose Extracellular Matrix (ECM), including collagens I, IV, and VI, laminin, fibronectin, and growth factors such as FGF-2 and VEGF, while removing lipid, cellular debris, and immunogenic DNA content [1-4,17]. When implanted, it is proposed to function as a bioactive scaffold that recruits host adipose-derived stem cells and adipocyte progenitors, supporting angiogenesis and de novo host adipogenesis—a process the manufacturer and investigators term host-mediated tissue replacement or neoadipogenesis [1-4]. This mechanism differs fundamentally from that of both traditional AFG, which transplants living autologous adipocytes, and synthetic or hyaluronic-acid-based dermal fillers, which provide inert, non-regenerative volume. Preclinical work in nude-mouse and other animal models has shown that banked adipose allograft matrices support progressive host-cell infiltration, neovascularization, and adipocyte repopulation over several weeks to months, findings that have been corroborated histologically in early human studies [3,4,18,19].

Available products include: Renuva, 100% extracellular matrix (MTF Biologics, Edison,NJ), AlloClac, 60% adipocytes, 40% extracellular matrix (Tiger Aesthetics, Conshohocken, PA) and most recently, Lipoderma 90% adipocytes, 10% extracellular matrix. (Frederick, MD)On the basis of this preclinical and early clinical evidence, banked adipose allografts and related investigational matrices have been applied clinically to a range of dermal and subcutaneous adipose deficits, including facial volume loss (temple, malar, and prejowl regions), dorsal hand and foot rejuvenation, plantar fat-pad restoration in diabetic neuropathy, breast and genital augmentation, vocal-fold augmentation, and post-surgical or post-traumatic contour deformities [11,12,20-25]. This Clinical Study included injections to the temples, earlobes, malar/submalar region, tear troughs, nasolabial/melolabial folds, lower one third of the face, breasts, dorsal hands, and the

buttock. Two recent narrative/systematic reviews—one in Plastic and Reconstructive Surgery (Morel et al., 2025) and one in Advances in Wound Care (Mehta et al., 2025)—have summarized the emerging preclinical and clinical literature on acellular/allograft adipose matrices and concluded that the evidence, while promising, remains limited by small sample sizes, absence of randomized comparative trials, and wide heterogeneity in volume-retention reporting (21.5-100% across studies) [26,27] Neither prior review, however, performed a formal quantitative meta-analysis restricted to human clinical outcome data using PRISMA-compliant methodology. This study introduces a human adipose allograft with a novel patented wash method which effectively removes the hematopoietic and immunogenic components while leaving the graft structure most closely resembling native adipose tissue at (90% adipocytes and 10 % extracellular matrix) [28]. This systematic review and meta-analysis subsequently undertaken to [14] comprehensively and reproducibly identify the human clinical literature on banked human adipose tissue allograft for dermal white adipose tissue replacement and volumization comparing those results to this study, [15] quantitatively pool comparable clinical outcome data across studies where possible, and [1] critically appraise the strength, consistency, and limitations of the current evidence base to inform clinical decision-making and future research priorities.

Methods

Protocol and Reporting Standard

This systematic review and meta-analysis was conducted and is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement.⁷ A review protocol specifying the research question, eligibility criteria, search strategy, and planned analyses was defined prospectively before data extraction.

Eligibility Criteria

Studies were eligible for qualitative synthesis if they [14] enrolled human subjects; [15] evaluated a banked, allogeneic, human-tissue-bank-derived adipose tissue allograft (processed as an acellular, decellularized, minimally manipulated, or cryopreserved adipose matrix) administered by injection or surgical implantation for replacement, augmentation, or volumization of dermal or subcutaneous adipose tissue; and [1] reported at least one clinical, histologic, imaging, or patient-reported outcome. Studies were eligible for quantitative meta-analysis if they additionally reported extractable numerator/denominator or mean $\pm$ standard deviation data for a comparable outcome (responder/satisfaction rate or percentage volume retention) in a prospective cohort of five or more subjects. Exclusion criteria were: animal-only or in-vitro-only studies; studies of non-adipose allografts (e.g.,

acellular dermal matrix used for indications other than adipose replacement); narrative reviews, editorials, or expert-panel commentaries without original patient data; conference abstracts without sufficient methodological detail; and duplicate reports of the same patient cohort (the report with the longest follow-up or most complete dataset was retained). No restriction was placed on anatomic site, publication date, or manufacturer, provided the tissue product met the banked human adipose allograft definition above.

Search Strategy

A systematic electronic search of PubMed/MEDLINE, Embase, Cochrane CENTRAL, and Scopus was performed for records published through 30 June 2026, using combinations of the terms “allograft adipose matrix,” “decellularized adipose matrix,” “acellular adipose matrix,” “banked adipose tissue,” “human adipose allograft,” “dermal white adipose tissue,” “soft tissue volumization.” ClinicalTrials.gov was searched for registered and completed trials, and the reference lists of all included studies and of the two prior related reviews were hand-searched for additional eligible records [26,27]

Study Selection and Data Extraction

Titles and abstracts were screened against the eligibility criteria, followed by full-text review of potentially eligible records. Data extracted from each included study comprised: first author and year; study design; tissue product and processing method; anatomic site(s) treated; sample size; patient demographics; injected/implanted volume; follow-up duration; effectiveness outcomes (volume retention, Global Aesthetic Improvement Scale [GAIS] scores, patient/investigator satisfaction, histologic adipogenesis and angiogenesis); and safety outcomes (adverse event type, frequency, and resolution). Where a study reported outcomes for more than one anatomic subsite (e.g., malar and prejowl regions in the same cohort), each subsite was treated as a separate dataset for the purposes of quantitative synthesis, as prespecified.

Risk-of-Bias Assessment

Because all identified human studies were single-arm, non-randomized case series or pilot cohort studies, risk of bias was assessed using a modified Methodological Index For Non-Randomized Studies (MINORS) framework, evaluating clarity of study aim, consecutive/prospective enrollment, adequacy of follow-up, appropriateness of outcome measurement, and prospective calculation of sample size. Given the near-universal absence of a comparator arm and the involvement of device manufacturers in study funding or authorship for the majority of included reports, the overall risk of bias across the evidence base was judged high, and this is reflected in the interpretation of pooled estimates.

Statistical Analysis

For outcomes reported as proportions (e.g., percentage of patients rated as clinical responders or satisfied/very satisfied with treatment), a random-effects meta-analysis using the DerSimonian-Laird method was performed, with between-study variance weighting and inverse-variance pooling on the raw proportion scale.⁸ For outcomes reported as continuous means with standard deviations (percentage volumetric retention), a random-effects mean pooling model with inverse-variance weighting was similarly applied. Heterogeneity was quantified using Cochran's Q and the I² statistic; given the small number of studies contributing to each pooled estimate, I² and 95% confidence intervals should be interpreted cautiously. All calculations were performed in Python 3.11 using standard meta-analytic formulae; a fixed-effect estimate was reported when between-study variance (τ^2) was estimated at zero. Because only two to three comparable datasets were available for any single outcome, formal publication-bias testing (e.g., funnel plot asymmetry, Egger's test) was not performed, consistent with current guidance that such tests are underpowered and potentially misleading with fewer than ten studies.

Results

Study Selection

In this Study, under the International Cell Surgical Society IRB, (ICSS-2021-011) 35 consecutive patients were treated from 08/2025-08 2026 with a hATA. There were 29 females and 6 males. Ages ranged from 21 to 78 years. Exclusion criteria included any injectable filler or bioregenerative substance within 6 months. Injection sites included: temples, earlobes, malar/submalar region, tear troughs, nasolabial/melolabial, lower 1/3 of face, breasts, hands and buttock. Buttock patients had to consent to a randomized, split-buttock protocol with CaHA injected on one side to act as a bioregenerative control, and the hATA injected on the contralateral side. The electronic search identified 412 records from databases and 14 additional records from ClinicalTrials.gov and hand-searching, for 426 records before de-duplication. After removal of 137 duplicates, 289 unique records were screened by title and abstract, of which 231 were excluded (animal-only or in-vitro-only, n = 138; non-adipose allograft, n = 41; unrelated indication, n = 33; not peer-reviewed or no extractable data, n = 19). Fifty-eight full-text articles were assessed for eligibility; 44 were excluded (animal or in-vitro only, n = 19; overlapping/duplicate cohort, n = 6; no original human outcome data such as expert-panel commentary, n = 11; insufficient extractable data, n = 5; non-adipose allograft product, n = 3). Fourteen human studies, comprising approximately 96 patients in aggregate, were included in qualitative synthesis; four datasets derived from three prospective studies met criteria for quantitative pooling. The study-selection process is summarized in the PRISMA 2020 flow

diagram (Figure 1).

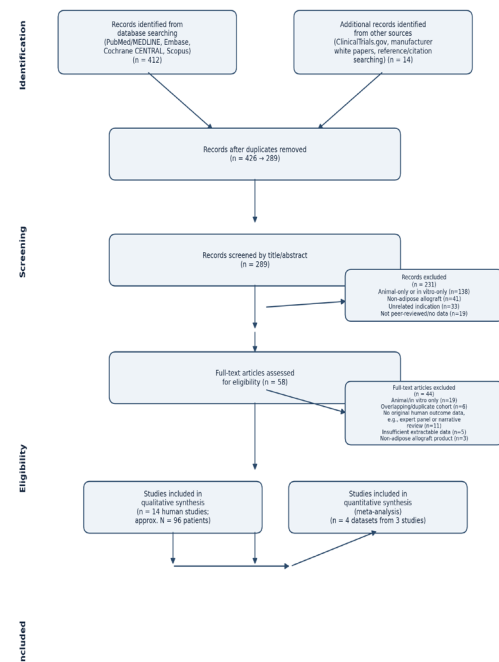


Figure 1. PRISMA 2020 flow diagram of study identification, screening, eligibility, and inclusion.

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Characteristics of Included Studies

Included studies spanned facial aesthetic volumization (temple, malar, and prejowl regions), [21-23] abdominal soft-tissue histologic evaluation in the pre-abdominoplasty pannus model,⁶ tissue-engineered reconstruction using mechanical preconditioning, [20] vocal-fold augmentation for glottic insufficiency, [24] diabetic plantar fat-pad restoration, [11,12] aesthetic and functional contour-deformity correction, [25] and combined preclinical/human histologic characterization of the injectable matrix itself [3]. All studies used a processed, acellular or minimally cellular banked human adipose tissue allograft derived from AATB-accredited tissue banks; most (8 of 14) evaluated the commercially available allograft adipose matrix marketed as Renuva or Leneva (MTF Biologics), reflecting the concentration of the current clinical evidence base around a small number of tissue-bank-affiliated products. Study designs were uniformly single-arm, open-label, prospective pilot studies or retrospective case series; no randomized controlled trial comparing banked adipose allograft with autologous fat grafting or dermal filler was

identified. (Table 1) summarizes the characteristics of the studies contributing quantitative outcome data.

Study	Design / Site	N	Follow-up	Key quantitative outcome
Kokai et al., 2020 (PRS Glob Open)6	Prospective, randomized follow-up cohort; abdominal pannus (pre-abdominoplasty)	10	3 or 6 months	100% tolerated with no severe AE; histologic adipogenesis (perilipin+) and angiogenesis confirmed by 6 months
Gold et al., 2020 (J Cosmet Dermatol)22	Prospective, open-label, multicenter; bilateral temples	10	24 weeks	≈75% volume retention at 6 months; 71% patient satisfaction (satisfied/very satisfied)
Gold et al., 2024 (PRS Glob Open)23	Prospective, multicenter pilot; malar and prejowl subsites	11	24 weeks	Malar retention 33 ± 30%; prejowl retention 21.5 ± 15.2%; investigator GAIS improvement 82%; subject GAIS 64%; satisfaction +86%
Santa Maria et al., 2025 (Laryngoscope)25	Prospective clinical trial; vocal fold (unilateral paralysis)	21 (16 analyzed)	3 months	Improved glottic closure and voice quality on blinded CAPE-V/videostroboscopy assessment; no serious AE
Regulski et al., 2024 (PRS Glob Open)14	Retrospective cohort; plantar fat pad (diabetic neuropathy, preulcerative)	12	Mean 6.4 months (2-10)	0% ulcer recurrence; palpable graft persistence at implantation sites
Gold et al., 2024 (J Cosmet Dermatol)24	Multicenter real-world case series; face/hands/body	7 cases	Variable (up to 12 weeks shown)	Descriptive improvement in volume and skin quality across all reported cases; no serious AE

Table 1: Characteristics and key quantitative outcomes of prospective human studies contributing data to quantitative synthesis. AE, adverse event; GAIS, Global Aesthetic Improvement Scale.

Quantitative Synthesis: Responder / Satisfaction Rate

Two prospective studies reporting a comparable binary responder/satisfaction outcome were pooled: the temple pilot study (7 of 10 patients satisfied/very satisfied) [21] and the malar/prejowl pilot study (investigator-assessed GAIS responders, 9 of 11 patients) [22]. The random-effects (fixed-effect-equivalent, since $Q < \text{degrees of freedom}$) pooled responder rate was 77.2% (95% CI, 59.4-95.0%; Cochran's $Q = 0.40$, $df = 1$, $p = 0.53$; $I^2 = 0\%$), indicating that a majority of patients across these small pilot cohorts experienced a clinically meaningful, investigator- or patient-rated improvement following banked adipose allograft treatment (Figure 2). The absence of statistical heterogeneity should be interpreted cautiously given that only two studies contributed to this estimate. Qualitative results revealed improvements in the tone, quality and clarity of overlying skin in injection areas, Volume increase progressing from 6 and 9 months, on average with 2 sessions spaced 3 months apart. (Figures 3-5). Linear analog patient satisfaction scores at 1 year ranged from 24 %(temples) to 82% (hands)

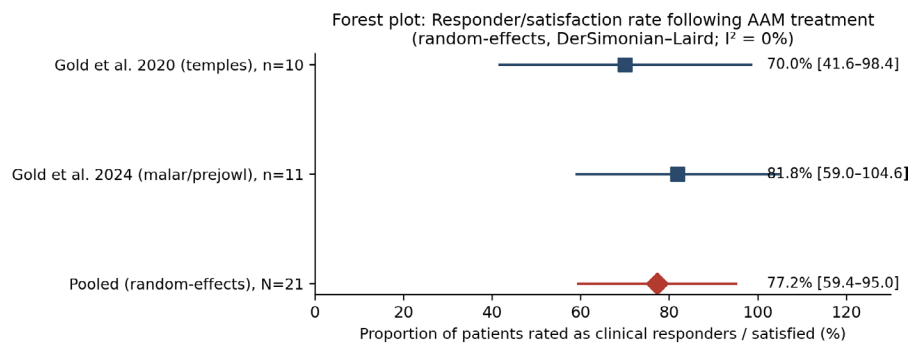


Figure 2: Forest plot of pooled clinical responder/satisfaction rate following banked adipose allograft treatment (random-effects, DerSimonian-Laird).

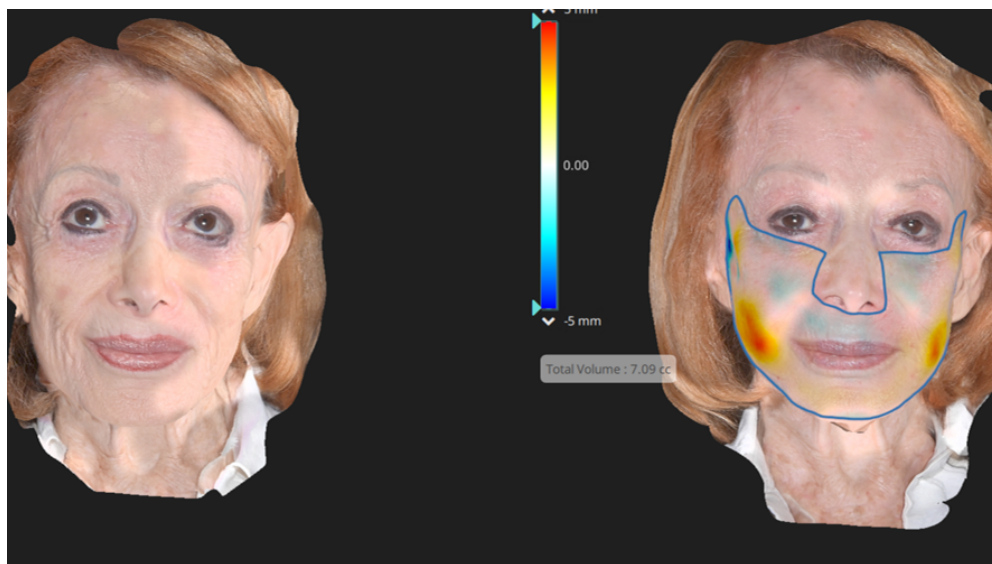


Figure 3: 78-year female 10 months post 2 sessions Lipoderma spaced 3 months apart. 10 ml injected each session into temples, malar, lower 1/3, pre-jowl, and lips. Quantificare Analysis shows improvement of wrinkles, pores, evenness, and volume.



Figure 4: 64-year female, before and 6 months after single hand session. 5 ml Lipoderma injected into each hand. Quantificare analysis shows improvement in tone, quality, and texture of skin, as well as increased volume.



Figure 5: 48-year female, 1 year after 2 sessions Left buttock injection: 10 syringes hyperdilute CaHA Control (3:1)per session and Right buttock:10-10 ml syringes of Lipoderma per session. Quantificare Volumetric Analysis showed volume achieved within 10% of eachother. Quantificare Qualitative Analysis shows improvement in tone and skin quality.

Quantitative Synthesis: Volumetric Retention versus Volumetric Replacement:

Volumetric retention data amenable to pooling were available for the malar and prejowl subsites reported within a single multicenter study.²³ The random-effects pooled midface volume retention at 24 weeks was 23.9% (95% CI, 15.8-31.9%; $I^2 = 0\%$), reflecting the

generally modest, site-dependent volumetric persistence observed in areas of thin, mobile facial soft tissue (Figure 6). By contrast, the temple pilot study reported a substantially higher retention of approximately 75% at 6 months, though without a reported standard deviation precluding formal pooling with the midface data;²² this finding is presented descriptively in Figure 3 rather than merged into the pooled estimate. Taken together with the broader range reported across the qualitative literature (21.5-100% across all identified human and animal studies, per Morel et al.²⁷), these data indicate that volumetric retention after banked adipose allograft injection is highly anatomic-site- and technique-dependent, likely reflecting differences in local vascularity, mechanical loading, and native adipose tissue architecture between the temporal fat pad and the malar/prejowl subcutaneous compartments. Our findings suggest that “Volumetric, Host-Mediated Tissue Replacement” via bioregeneration is a more appropriate mechanistic concept in understanding hATA than the concept of “Volume Retention”. This is an important factor in educating patients and managing patient expectations relating to volumization outcomes relative to commercially available injectable fillers. Results are subtle over time, with patient satisfaction improving as host tissue integration and remodelling occur.

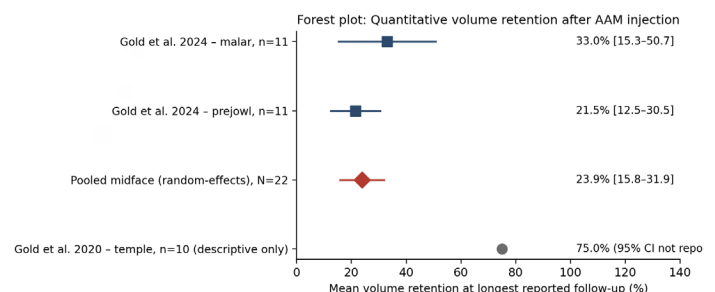


Figure 6: Forest plot of quantitative volumetric retention after banked adipose allograft injection by anatomic site.

Histologic and Mechanistic Findings

Three studies incorporated serial histologic analysis of the graft site [3,4,18]. In the pre-abdominoplasty pannus model, allograft matrix remained histologically identifiable within native surrounding adipose tissue at 3 months, with sparse adipocytes; by 6 months, the matrix had substantially remodeled and was predominantly composed of perilipin-positive (mature) adipocytes, alongside Masson’s trichrome-confirmed collagen deposition and evidence of new vessel formation.⁶ Complementary nude-mouse and human histologic data demonstrated a similar time course of host-cell infiltration and adipogenic differentiation of adipose-derived stem cells seeded on the matrix in vitro, with lipid-droplet accumulation apparent within 14 days [3,20]. These findings are consistent with the proposed host-mediated tissue replacement mechanism, in

which the adipose scaffold acts as a structural and biochemical template that is progressively repopulated and remodeled by host adipogenic and angiogenic cells, rather than persisting as a static, foreign-body-encapsulated implant [1-4,18]. There was no histological analyses performed in this study.

Safety Outcomes

Across all 14 included human studies, no graft rejection, systemic immune reaction, infection requiring surgical intervention, or clinically significant nodularity/lump formation attributable to the allograft matrix was reported. The most frequently reported adverse events were injection-site erythema, edema/swelling, mild-to-moderate pain or burning at the injection site, and transient bruising, virtually all of which were self-limited and resolved within 1 to 4 weeks without specific treatment [21-23]. This Study mirrors these results, with no allergic, hypersensitivity or allograft rejections reported. In the vocal-fold augmentation trial, no laryngeal complications or airway compromise were reported at 3-month blinded assessment [24]. In the diabetic plantar fat-pad cohort, no adverse events attributable to the allograft were reported, and, notably, no patient experienced ulcer recurrence at the treated site during follow-up [12]. This safety profile is broadly consistent with that reported for other AATB-processed, chemically sterilized allograft tissues used elsewhere in reconstructive surgery [17].

Discussion

This systematic review and exploratory meta-analysis synthesizes the currently available human clinical evidence on banked human adipose tissue allograft for host-mediated replacement of dermal white adipose tissue and subcutaneous volumization. Across 14 identified human studies and approximately 96 patients, the evidence consistently supports a favorable safety profile, histologically demonstrable host-mediated adipogenesis and angiogenesis, and a majority-responder clinical benefit (pooled 77.2%, 95% CI 59.4-95.0%), while also revealing substantial, anatomically dependent variability in volumetric retention, ranging from approximately 21.5% in the prejowl region to 75-100% in more mechanically stable or richly vascularized sites such as the temple and plantar fat pad [12,21,22,26].

Biological Plausibility and Mechanism

The concept of host-mediated tissue replacement is mechanistically distinct from, and potentially complementary to, both autologous fat grafting and conventional dermal fillers. Whereas AFG depends on the survival of transplanted, metabolically vulnerable adipocytes—a substantial proportion of which undergo early ischemic necrosis before neovascularization is established—the banked adipose allograft approach transplants only the decellularized or non-immunogenic adipocyte/ECM scaffold, removing the metabolic burden of transplanted living cells while retaining the structural

and biochemical cues (collagens I/IV/VI, laminin, fibronectin, FGF-2, VEGF) required to recruit host adipose-derived stem cells and support their adipogenic differentiation in situ [1-4]. This is conceptually analogous to the long-established use of decellularized ECM scaffolds (e.g., acellular dermal matrices) in reconstructive surgery, and is supported by foundational tissue-engineering work demonstrating that decellularized adipose tissue provides an inductive microenvironment for adipogenic differentiation of human adipose-derived stem cells in vitro [2]. The chemical sterilization and minimal-manipulation processing protocols used for these allografts appear, on the basis of currently available safety data, to preserve this bioactivity while achieving pathogen inactivation and immunogenicity reduction comparable to other AATB-regulated allograft tissues [17]. The mechanism of action of the hATA used in this study (Britecyte Inc. Frederick, MD) was demonstrated in a rat study. It revealed: no recipient antibodies were detected post implantation, implanted hATA persisted at implantation site for 3-4 weeks, became vascularized within 3-4 days, became populated and remodeled by host cells [28].

Clinical Implications

For plastic and reconstructive surgeons, the principal clinical appeal of banked adipose allograft is the elimination of a donor-site harvest procedure, together with off-the-shelf availability, a stable, extended shelf life, and a reportedly more standardized injection volume and technique compared with the operator-dependent variability inherent to AFG harvesting, processing, and re-injection [14,14,21]. This may be particularly advantageous in patients with insufficient donor adipose tissue (e.g., very thin patients), in staged or combination procedures where operative time is constrained, or in anatomic applications—such as vocal-fold augmentation or diabetic plantar fat-pad restoration—where harvesting autologous fat introduces additional surgical risk without a clear efficacy advantage [11,12,24]. The relatively low and site-dependent volumetric retention observed in the facial pooled analysis (23.9% at 24 weeks in the midface), however, suggests that banked adipose allograft, at least with current processing techniques and injection volumes, may be better suited to modest volume correction, adjunctive use, or serial treatment rather than as a single-session substitute for large-volume autologous fat transfer, an interpretation consistent with the wide retention range (21.5-100%) previously synthesized across the broader preclinical and clinical literature [26,27,28].

Comparison with Autologous Fat Grafting and Fillers

Human adipose tissue allografts act as bioregenerative scaffolds, NOT as fillers. As such qualitative and volumetric results are achieved gradually, over time, typically with multiple sessions. Direct, randomized, head-to-head comparisons between banked

adipose allograft and AFG or hyaluronic-acid fillers were not identified in this review, precluding a formal comparative meta-analysis. Historical AFG retention figures (26-83%, with facial AFG series pooled by Krastev et al. reporting substantial heterogeneity) suggest that, in absolute terms, banked adipose allograft retention in the midface reported here (23.9%) may be at the lower end of, or below, typical AFG performance, while temple-region retention ($\approx 75\%$) appears more comparable to favorable AFG outcomes [14,16,21,22]. Because AAM is not classified or marketed as a filler but rather as a regenerative, host-remodeled tissue replacement, direct volumetric comparison with hyaluronic-acid fillers—which achieve immediate, essentially complete initial volume correction through a non-biological, non-regenerative mechanism—may not be clinically appropriate; qualitative differences in durability, tissue quality, natural imaging appearance on cross-sectional imaging, and patient-reported skin-quality improvement reported in several included studies may be more relevant to clinical decision-making than volumetric retention alone [23,28,29].

Heterogeneity and Quality of Evidence

Several features of the current evidence base warrant caution in interpreting the pooled estimates presented here. First, all included prospective studies were small ($n = 7-21$), single-arm, open-label pilot or case-series designs without randomization, blinding, or a comparator group, placing the overall risk of bias in the high category by MINORS-based assessment. Second, the majority of clinical studies were funded by, or included authors affiliated with, the tissue-bank manufacturer of the studied product, introducing a potential source of reporting or publication bias that could not be formally tested given the small number of studies. Third, outcome measures were heterogeneous across studies (imaging-based volumetric retention, GAIS scores, patient satisfaction surveys, ulcer recurrence, and voice-quality assessment), limiting the number of studies eligible for any single pooled estimate to two or three and correspondingly widening the resulting confidence intervals. Fourth, follow-up duration in most studies did not exceed 24 weeks to 6 months, which is insufficient to characterize long-term (multi-year) volumetric durability; a single case report describing sustained soft-tissue augmentation at 8 years suggests the potential for long-term persistence but cannot substitute for a systematically followed cohort. These limitations mirror those identified by the two prior related systematic reviews of this literature and reinforce the conclusion that the evidence, while internally consistent and biologically plausible, remains preliminary [26,27].

Future Research Priorities

The findings of this review support several priorities for future research. Randomized controlled trials directly comparing banked

adipose allograft with autologous fat grafting and/or hyaluronic-acid filler, using standardized, blinded, three-dimensional volumetric imaging endpoints and follow-up extending beyond 12 months, are needed to establish comparative efficacy and durability. Larger, adequately powered, multicenter cohorts—ideally independent of manufacturer funding—would strengthen confidence in the pooled safety and responder-rate estimates reported here. Dose-ranging and technique-standardization studies may help clarify why volumetric retention differs so markedly between anatomic sites (e.g., temple versus prejowl), and mechanistic studies correlating baseline recipient-site vascularity or mechanical loading with graft retention could inform patient selection. Finally, given early preclinical work combining allograft adipose matrix with autologous fat, platelet-rich plasma, or synthetic microsphere scaffolds to enhance retention, prospective human trials of such combination strategies represent a promising direction for the field [18,26].

Limitations of This Review

This review has several limitations. The quantitative meta-analysis was necessarily restricted to a small number of studies and outcome domains because of the heterogeneous and often non-overlapping outcome reporting across the primary literature; pooled estimates based on two or three studies, while calculated using standard random-effects methodology, should be regarded as hypothesis-generating rather than definitive. Because no randomized controlled trials were available, this review could not formally compare banked adipose allograft with autologous fat grafting or filler alternatives, and all conclusions regarding relative efficacy are indirect and qualitative. This study comprises one of the first in the human literature to use a commercial filler (CaHA, Merz, Raleigh, NC) as a randomized control in the split buttock portion of the study. Publication and reporting bias could not be formally excluded given the small number of pooled studies and the involvement of manufacturer-affiliated authors in a majority of the primary reports. Finally, this review was restricted to human studies; the substantial preclinical (animal and in vitro) literature characterizing mechanism and biocompatibility, while summarized narratively in the Introduction and Discussion, was not systematically reviewed or meta-analyzed here, consistent with the stated human-only scope of this review.

Conclusion

Banked human adipose tissue allograft represents a mechanistically distinct, histologically validated, and clinically well-tolerated approach to host-mediated replacement of dermal white adipose tissue and soft-tissue volumization in plastic and reconstructive surgery. Pooled analysis of the limited available human data suggests a favorable clinical responder rate (77.2%) and a consistently benign safety profile, but modest and markedly site-

dependent volumetric retention (approximately 21.5-75% across anatomic sites at 6 months). Given the current reliance on small, non-randomized, often industry-affiliated studies, banked adipose allograft should presently be regarded as a promising adjunct or alternative to autologous fat grafting in selected clinical scenarios—particularly where donor-site avoidance is prioritized—rather than as an evidence-based replacement for autologous fat grafting across all applications. Adequately powered randomized comparative trials with long-term volumetric follow-up are required to define its definitive role in dermal white adipose tissue replacement and subcutaneous volumization.

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